

$$A_{i_0} \subseteq \bigcup_{i \in I} A_i \Rightarrow \left(\bigcup_{i \in I} A_i \right)' = \bigcap_{i \in I} A_i' \subseteq A_{i_0}'$$

Since A_{i_0}' is finite &

$$\left(\bigcup_{i \in I} A_i \right)' \subseteq A_{i_0}', \quad \left(\bigcup_{i \in I} A_i \right)' \text{ is finite}$$

$$\Rightarrow \bigcup_{i \in I} A_i \in \mathcal{T}.$$

$\therefore (X, \mathcal{T})$ is a topological space.

We already know that $\emptyset, X \in \mathcal{T}_K$.

Let $A \subsetneq X$ s.t. $A \neq \emptyset$.

A is $\mathcal{T}$ -closed iff $A' \in \mathcal{T}$

$A' \in \mathcal{T} \iff A = (A')'$ is finite

In other words if $\emptyset \neq A \subsetneq X$

then A is $\mathcal{C}$ -closed $\Leftrightarrow A$ is finite, ^{||}

5. Let $X = \{a, b, c\}$. Consider the following classes of subsets of X :

$$\mathcal{T}_1 = \{\emptyset, \{a\}, \{a, b\}, X\}$$

$$\mathcal{T}_2 = \{\emptyset, \{b\}, \{a, b\}, \{b, c\}, X\}$$

$$\mathcal{T}_3 = \{\emptyset, \{c\}, \{a, c\}, \{b, c\}, X\}$$

$$\mathcal{T}_4 = \{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$$

$$\mathcal{T}_5 = \{\emptyset, \{a\}, \{b\}, \{a, c\}, X\}.$$

Determine if $\mathcal{T}_i$ is a topology on X for $i = 1, 2, 3, 4, 5$. If $\mathcal{T}_i$ is a topology on X find all $\mathcal{T}_i$ -closed subsets of X .

Solution:

$\mathcal{T}_1$ is a topology on X .

$\mathcal{T}_2$ is also a topology on X .

Similarly you can show that $\mathcal{T}_3$ & $\mathcal{T}_4$ are topologies on X .

$\{a\}, \{b\} \in \mathcal{T}_5$. But $\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{T}_5$.

$\therefore \mathcal{T}_5$ is not a topology on X .

Let us find $\mathcal{T}_i$ -closed subsets of X
for $i = 1, 2, 3, 4$:

$$\mathcal{T}_1 = \{ \emptyset, \{a\}, \{a, b\}, X \}$$

$$(\mathcal{T}_1)_K = \{ X, \{b, c\}, \{c\}, \emptyset \}$$

$$\mathcal{T}_2 = \{ \emptyset, \{b\}, \{a, b\}, \{b, c\}, X \}$$

$$(\mathcal{T}_2)_K = \{ X, \{a, c\}, \{c\}, \{a\}, \emptyset \}$$

$$\mathcal{T}_3 = \{ \emptyset, \{c\}, \{a, c\}, \{b, c\}, X \}$$

$$(\mathcal{T}_3)_K = \{ X, \{a, b\}, \{b\}, \{a\}, \emptyset \}$$

$$\mathcal{T}_4 = \{ \emptyset, \{a\}, \{b\}, \{a, b\}, X \}$$

$$(\mathcal{T}_4)_K = \{ X, \{b, c\}, \{a, c\}, \{c\}, \emptyset \}$$

6. We know that the intersection of finitely many open sets is open. But if we consider an arbitrary collection of open sets their intersection need not be open.

Let $X = \mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$ &

$\mathcal{C}$ be the cofinite topology on X .

Consider $A_n = \{1, n, n+1, n+2, n+3, \dots\}$

for $n = 1, 2, 3, \dots$. Show that

$A_n \in \mathcal{C}$ for each $n = 1, 2, \dots$. Find

$\bigcap_{n=1}^{\infty} A_n$. Is $\bigcap_{n=1}^{\infty} A_n$ $\mathcal{C}$ -open?

Solution:

$$A_1 = \{1, 1, 2, 3, 4, \dots\}$$

$$= \{1, 2, 3, 4, \dots\} = \mathbb{Z}^+ = X.$$

$X \setminus A_1 = X \setminus X = \emptyset$ is a finite set.

$$A_2 = \{1, 2, 3, 4, \dots\} = \mathbb{Z}^+ = X.$$

$X \setminus A_2 = \emptyset$ is a finite set.

$\therefore A_1, A_2 \in \mathcal{C}$.

$$X \setminus A_n = \{2, 3, 4, \dots, n-1\} \text{ for } n \geq 3.$$

Because for $n \geq 3$ $X \setminus A_n$ is finite

$A_n \in \mathcal{C}$.

$$\bigcap_{n=1}^{\infty} A_n = A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5 \cap \dots$$

$$= \{1, 2, 3, 4, 5, \dots\} \cap \{1, 2, 3, 4, 5, \dots\} \\ \cap \{1, 3, 4, 5, 6, \dots\} \cap \{1, 4, 5, 6, \dots\} \\ \cap \{1, 5, 6, 7, \dots\} \cap \dots \\ = \{1\}$$

$$\left[\bigcap_{n=1}^{\infty} A_n \right]' = \{1\}' = \mathbb{Z}^+ \setminus \{1\} \\ = \{2, 3, 4, \dots\} \text{ is}$$

not a finite set.

$$\therefore \bigcap_{n=1}^{\infty} A_n \notin \mathcal{C}.$$

7. Let $\{\tau_i\}_{i \in I}$ be any collection of topologies on a (nonempty) set X . Then show that $\tau = \bigcap_{i \in I} \tau_i$ is a topology on X .

Solution:

(T1) Because of (T1) $\emptyset, X \in \tau_i$ for all $i \in I$. This implies that $\emptyset, X \in \bigcap_{i \in I} \tau_i = \tau$.

(T2) Let $A_1, A_2 \in \tau$. Then $A_1, A_2 \in \tau_i$ for every $i \in I$. $A_1 \cap A_2 \in \tau_i$ for every $i \in I$.
 $\Rightarrow A_1 \cap A_2 \in \tau$.

(T3) Let $A_j \in \tau$ for $j \in J$. Thus $A_j \in \tau_i$ for every $j \in J$ & every $i \in I$. Because τ_i satisfies (T3)

$\bigcup_{j \in J} A_j \in \tau_i$ for each $i \in I$. Thus

$$\bigcup_{j \in J} A_j \in \mathcal{T}.$$

$\therefore \mathcal{T}$ is a topology on X .

8. Let $(X, \mathcal{T}_1)$ & $(X, \mathcal{T}_2)$ be topological spaces & $\mathcal{T} = \mathcal{T}_1 \cup \mathcal{T}_2$.

Can we always say that $\mathcal{T}$ is a topology on X ? NO! Not always.

Counterexample: Let $X = \{a, b, c\}$,
 $\mathcal{T}_1 = \{\emptyset, \{a\}, X\}$ & $\mathcal{T}_2 = \{\emptyset, \{b\}, X\}$.

Then $\mathcal{T}_1$ & $\mathcal{T}_2$ are topologies on X
& $\mathcal{T} = \mathcal{T}_1 \cup \mathcal{T}_2 = \{\emptyset, \{a\}, \{b\}, X\}$.

But although $\{a\}, \{b\} \in \mathcal{T}$,
 $\{a\} \cup \{b\} = \{a, b\} \notin \mathcal{T}$. Thus $\mathcal{T}$ is
not a topology on X .

Coarser and finer topologies

Let $\mathcal{T}_1$ & $\mathcal{T}_2$ be two topologies on a nonempty set X . Assume that every $\mathcal{T}_1$ -open subset of X is also a $\mathcal{T}_2$ -open subset of X . In other words $\mathcal{T}_1$ is a subclass of $\mathcal{T}_2$, i.e. $\mathcal{T}_1 \subseteq \mathcal{T}_2$. In this case we say that $\mathcal{T}_1$ is coarser or smaller (or weaker) than $\mathcal{T}_2$ OR that $\mathcal{T}_2$ is finer or larger than $\mathcal{T}_1$.

Observe that the collection $\{\mathcal{T}_i\}_{i \in I}$ of all topologies on X is partially ordered by class inclusion. We say that two topologies on X are not comparable if neither is coarser than the other.

Examples :

1. Let $X = \{a, b, c, d, e\}$. Then show that the following 2 topologies on X are not comparable:

$$\tau_1 = \{\emptyset, X, \{a\}, \{c, d\}, \{a, c, d\}, \{b, c, d, e\}\}$$

$$\tau_2 = \{\emptyset, X, \{a\}, \{b\}, \{a, b\}\}.$$

Solution:

$$\{c, d\} \in \tau_1 \quad \text{but} \quad \{c, d\} \notin \tau_2 \Rightarrow \tau_1 \not\subseteq \tau_2$$

$$\text{Similarly } \{b\} \in \tau_2 \quad \text{but} \quad \{b\} \notin \tau_1$$

$$\Rightarrow \tau_2 \not\subseteq \tau_1.$$

$\therefore \tau_1$ & τ_2 are not comparable.

2. Let $X \neq \emptyset$. What are the coarsest & finest topologies on X ?

Solution: Let

$$\tau_1 = \{\emptyset, X\} \quad \& \quad \tau_2 = \mathcal{P}(X) = \mathcal{D}$$

be the indiscrete & discrete

topologies on X . τ_1 is the coarsest topology on X while τ_2 is the finest topology on X .

(This tells us that if τ is an arbitrary topology on X then $\tau_1 \subseteq \tau \subseteq \tau_2$.)

Proposition: Let $X \neq \emptyset$ & $\{\tau_i\}_{i \in I}$

a collection of topologies on X . Then we have the following:

(i) $\tau = \bigcap_{i \in I} \tau_i$ is coarser than

τ_i for each $i \in I$.

(ii) If $\tau = \bigcup_{i \in I} \tau_i$ is a topology

on X then it is finer than τ_i for each i .

Continuity and topological equivalence

Definition: Let (X, τ) and (Y, ϑ) be two topological spaces & $f: X \rightarrow Y$ be a function. We say that $f: X \rightarrow Y$ or $f: (X, \tau) \rightarrow (Y, \vartheta)$ is continuous relative to τ and ϑ or τ - ϑ continuous or simply continuous iff the inverse image $f^{-1}(V)$ of every ϑ -open subset V of Y is a τ -open subset of X , i.e. f is τ - ϑ continuous $\Leftrightarrow V \in \vartheta$ implies $f^{-1}(V) \in \tau$.

Theorem: Let (X, τ) , (Y, ϑ) and $(Z, \mathcal{U})$ be topological spaces, $f: X \rightarrow Y$ & $g: Y \rightarrow Z$ be continuous functions. Then the composite function $g \circ f: X \rightarrow Z$ is continuous.

Proof: Let $U \in \mathcal{U}$.

Because $g: (Y, \vartheta) \rightarrow (Z, \mathcal{U})$ is ϑ - $\mathcal{U}$